

## EJIN QI, China

WMO: 522670

Lat: 41.950N

Lon: 101.067E

Elev: 941

StdP: 90.52

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -20.9      | -18.5 | -31.3                         | 0.2 | -12.4 | -29.4 | 0.3 | -11.1 | 8.6                   | -5.8 | 7.5 | -6.1 | 1.4                   | 90   | 0.525 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.3                   | 37.5            | 17.3 | 36.0 | 16.7 | 34.5 | 16.2 | 19.3                | 30.7 | 18.4 | 30.2 | 17.7 | 30.0 | 3.1                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.2                            | 12.9 | 22.1 | 14.7 | 11.7 | 23.0 | 13.1 | 10.5 | 23.8 | 59.4          | 31.1 | 56.2 | 30.2 | 53.7 | 29.8 | 23.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 8.7               | 7.4  | 6.5 | -23.5                      | 39.8 | 3.0                | 1.3 | -25.6                                              | 40.8 | -27.4      | 41.5 | -29.0      | 42.3 | -31.2      | 43.2 |  |  |
|                   |      |     | -24.0                      | 20.6 | 2.8                | 1.4 | -26.0                                              | 21.7 | -27.6      | 22.5 | -29.2      | 23.3 | -31.3      | 24.3 |  |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.4  | -9.9  | -4.1  | 4.0   | 13.0  | 19.5  | 25.3  | 28.0  | 25.8  | 19.2  | 10.2  | 0.3   | -8.0 | (6)  |
| (7)  |                                                                                  | DBStd       | 13.69 | 4.06  | 4.65  | 5.23  | 4.86  | 4.06  | 3.11  | 2.92  | 3.03  | 3.86  | 4.43  | 4.67  | 4.37 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2134  | 616   | 395   | 195   | 26    | 1     | 0     | 0     | 1     | 54    | 291   | 557   |      | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3797  | 874   | 628   | 445   | 168   | 36    | 1     | 0     | 1     | 36    | 254   | 540   | 815  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2262  | 0     | 0     | 8     | 117   | 297   | 459   | 557   | 489   | 276   | 59    | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 883   | 0     | 0     | 0     | 8     | 73    | 209   | 299   | 232   | 62    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 10867 | 0     | 0     | 0     | 126   | 899   | 2579  | 3924  | 2692  | 625   | 21    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 5460  | 0     | 0     | 0     | 26    | 335   | 1313  | 2217  | 1364  | 204   | 2     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.9   | 2.4   | 2.6   | 3.2   | 3.6   | 3.4   | 3.1   | 2.9   | 2.8   | 2.6   | 2.6   | 2.7   | 2.5  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 31    | 0     | 0     | 1     | 1     | 3     | 4     | 10    | 7     | 3     | 2     | 1     | 0    | (15) |
| (16) |                                                                                  | PrecMax     | 79    | 2     | 6     | 5     | 9     | 38    | 12    | 47    | 21    | 23    | 17    | 7     | 2    | (16) |
| (17) |                                                                                  | PrecMin     | 8     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 17    | 0     | 1     | 1     | 2     | 7     | 3     | 10    | 7     | 5     | 4     | 1     | 0    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 3.8   | 10.7  | 21.3  | 29.5  | 34.6  | 38.2  | 40.2  | 38.2  | 33.6  | 25.7  | 15.6  | 5.0  | (19) |
| (20) |                                                                                  | MCWB        | -1.9  | 2.2   | 7.7   | 11.9  | 14.0  | 16.6  | 18.6  | 17.8  | 15.8  | 11.1  | 5.6   | -1.0  |      | (20) |
| (21) |                                                                                  | 2%          | DB    | 0.9   | 8.0   | 18.4  | 26.6  | 32.2  | 36.2  | 38.3  | 36.1  | 30.8  | 22.9  | 12.7  | 2.4  | (21) |
| (22) |                                                                                  | MCWB        | -3.7  | 0.6   | 6.1   | 10.3  | 13.4  | 15.8  | 17.7  | 17.1  | 14.2  | 9.8   | 4.1   | -2.6  |      | (22) |
| (23) |                                                                                  | 5%          | DB    | -1.2  | 6.1   | 15.7  | 24.4  | 30.3  | 34.7  | 36.7  | 34.5  | 29.0  | 20.6  | 10.5  | 0.6  | (23) |
| (24) |                                                                                  | MCWB        | -5.1  | -0.5  | 4.8   | 9.4   | 12.5  | 15.4  | 17.4  | 16.6  | 13.5  | 8.6   | 2.9   | -3.7  |      | (24) |
| (25) |                                                                                  | 10%         | DB    | -3.0  | 4.1   | 13.3  | 22.2  | 28.3  | 33.1  | 35.2  | 33.0  | 27.1  | 18.4  | 8.1   | -1.1 | (25) |
| (26) |                                                                                  | MCWB        | -6.5  | -1.9  | 3.6   | 8.4   | 11.6  | 14.9  | 16.9  | 16.2  | 12.9  | 7.6   | 1.5   | -4.8  |      | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | -1.7  | 2.5   | 8.1   | 12.5  | 15.1  | 18.3  | 21.2  | 20.7  | 18.0  | 12.0  | 6.0   | -0.8 | (27) |
| (28) |                                                                                  | MDCB        | 3.6   | 9.7   | 19.1  | 27.4  | 31.4  | 29.7  | 35.0  | 31.3  | 25.3  | 23.1  | 15.0  | 4.8   |      | (28) |
| (29) |                                                                                  | 2%          | WB    | -3.6  | 0.8   | 6.5   | 10.9  | 13.8  | 17.4  | 19.6  | 19.2  | 16.0  | 10.4  | 4.2   | -2.4 | (29) |
| (30) |                                                                                  | MDCB        | 0.7   | 7.4   | 17.3  | 24.7  | 30.1  | 30.2  | 32.0  | 29.2  | 27.8  | 21.4  | 12.1  | 2.1   |      | (30) |
| (31) |                                                                                  | 5%          | WB    | -5.0  | -0.4  | 5.1   | 9.7   | 13.0  | 16.7  | 18.8  | 18.2  | 14.6  | 9.1   | 3.0   | -3.6 | (31) |
| (32) |                                                                                  | MDCB        | -1.4  | 6.0   | 15.0  | 23.3  | 28.8  | 30.6  | 31.1  | 29.1  | 25.8  | 19.5  | 10.3  | 0.4   |      | (32) |
| (33) |                                                                                  | 10%         | WB    | -6.4  | -1.8  | 3.8   | 8.6   | 12.1  | 15.9  | 18.1  | 17.3  | 13.6  | 7.9   | 1.6   | -4.8 | (33) |
| (34) |                                                                                  | MDCB        | -3.1  | 4.0   | 13.0  | 21.5  | 27.0  | 30.0  | 31.0  | 29.5  | 25.3  | 17.5  | 7.9   | -1.1  |      | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 11.8  | 13.2  | 13.6  | 13.4  | 13.3  | 12.9  | 12.3  | 12.5  | 13.2  | 13.6  | 12.0  | 11.1  |      | (35) |
| (36) |                                                                                  | 5% DB       | MCDBR | 12.8  | 14.9  | 16.0  | 16.0  | 15.8  | 15.0  | 14.0  | 14.2  | 15.0  | 15.8  | 14.3  | 11.8 | (36) |
| (37) |                                                                                  | MCWBR       | 9.4   | 9.6   | 8.6   | 7.7   | 6.8   | 5.4   | 4.5   | 4.9   | 5.8   | 7.7   | 8.6   | 8.3   |      | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 12.8  | 14.6  | 15.1  | 15.0  | 14.2  | 12.7  | 11.8  | 12.1  | 13.0  | 15.2  | 14.1  | 11.6 | (38) |
| (39) |                                                                                  | MCWBR       | 9.4   | 9.4   | 8.4   | 7.3   | 6.2   | 4.6   | 4.2   | 4.3   | 5.5   | 7.5   | 8.6   | 8.2   |      | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.310 | 0.359 | 0.401 | 0.446 | 0.458 | 0.475 | 0.471 | 0.449 | 0.417 | 0.367 | 0.331 | 0.311 |      | (40) |
| (41) |                                                                                  | taud        | 2.165 | 2.054 | 1.958 | 1.888 | 1.920 | 1.949 | 2.008 | 2.048 | 2.071 | 2.136 | 2.165 | 2.154 |      | (41) |
| (42) |                                                                                  | Ebn at Noon | 812   | 820   | 830   | 821   | 822   | 807   | 806   | 810   | 806   | 802   | 776   | 770   |      | (42) |
| (43) |                                                                                  | Edh at Noon | 99    | 131   | 163   | 189   | 188   | 183   | 171   | 159   | 145   | 119   | 99    | 92    |      | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.49  | 3.56  | 4.97  | 6.24  | 7.16  | 7.17  | 6.79  | 6.22  | 5.30  | 4.09  | 2.79  | 2.21  |      | (44) |
| (45) |                                                                                  | RadStd      | 0.17  | 0.18  | 0.29  | 0.26  | 0.32  | 0.42  | 0.33  | 0.30  | 0.24  | 0.11  | 0.13  | 0.16  |      | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +90     | +56     |
| (47) | Regional (2 neighbors) | +0.34 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +90     | +56     |

Nomenclature: See separate page